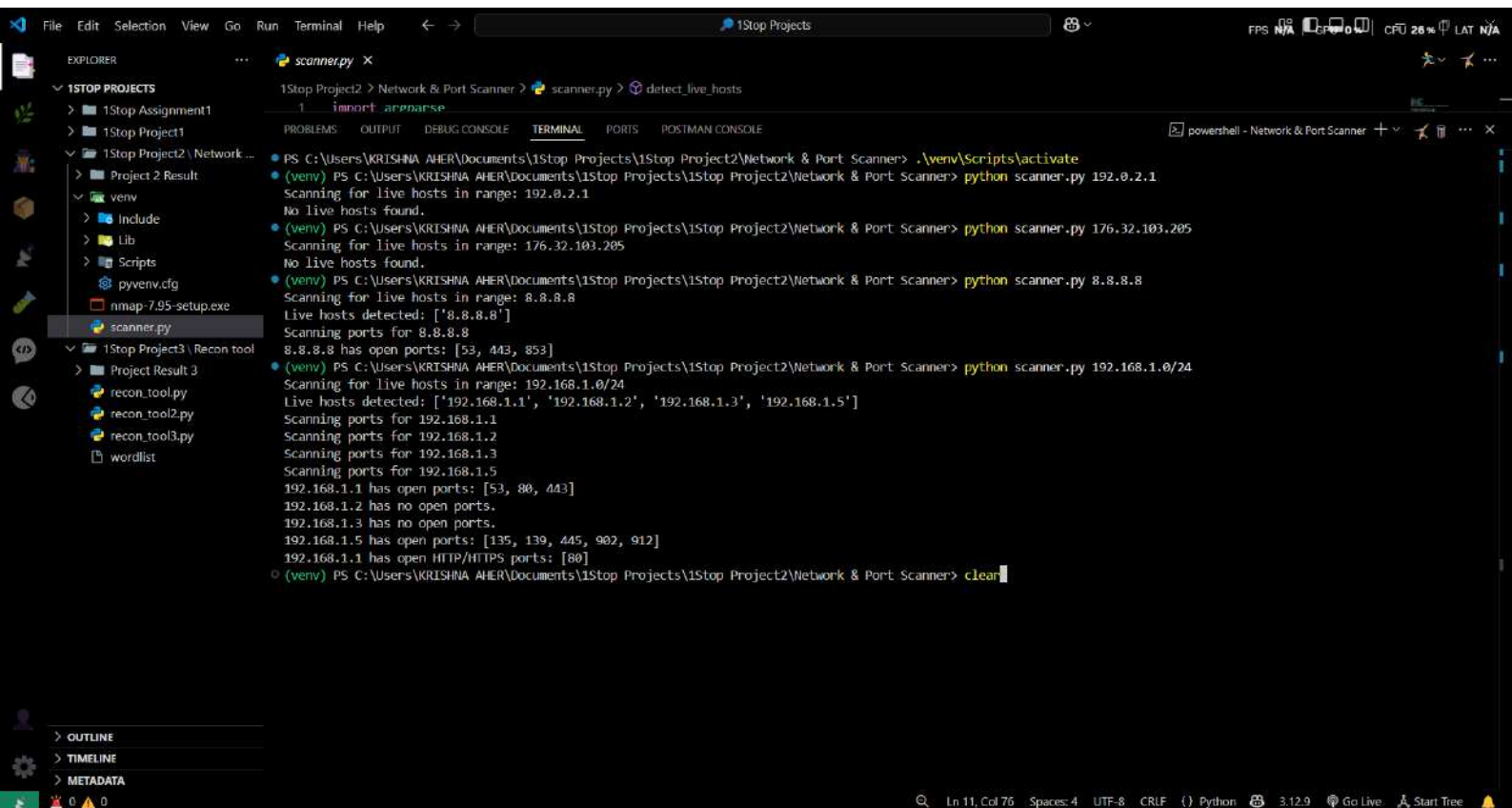


```

1 import argparse
2 import nmap
3 import socket
4 import requests
5 from concurrent.futures import ThreadPoolExecutor
6
7 # Function to detect live hosts
8 def detect_live_hosts(ip_range):
9     nm = nmap.PortScanner()
10    print(f"Scanning for live hosts in range: {ip_range}")
11    nm.scan(hosts=ip_range, arguments='-sn') # -sn flag for host discovery
12    live_hosts = []
13    for host in nm.all_hosts():
14        if nm[host].state() == 'up':
15            live_hosts.append(host)
16    return live_hosts
17
18 # Function to check open ports on a host
19 def check_ports(host):
20    open_ports = []
21    print(f"Scanning ports for {host}")
22    for port in range(20, 1025): # Scanning ports from 20 to 1024
23        with socket.socket(socket.AF_INET, socket.SOCK_STREAM) as s:
24            s.settimeout(1) # 1 second timeout for each port
25            result = s.connect_ex((host, port))
26            if result == 0:
27                open_ports.append(port)
28    return open_ports
29
30 # Function to check if a port is open on a host using requests (only for HTTP/HTTPS)
31 def check_http_ports(host):
32    http_ports = []
33    for port in [80, 443]: # Checking common HTTP/HTTPS ports
34        url = f"http://{host}:{port}"
35        try:
36            response = requests.get(url, timeout=1)
37            if response.status_code == 200:
38                http_ports.append(port)
39        except requests.exceptions.RequestException:
40            continue
41    return http_ports
42
43 # Function to run port scans concurrently
44 def scan_ports_concurrently(live_hosts):
45    with ThreadPoolExecutor(max_workers=5) as executor:
46        results = executor.map(check_ports, live_hosts)
47    return dict(zip(live_hosts, results))
48
49 # Main function to parse arguments and run the scans
50 def main():
51    # Parse command line arguments
52    parser = argparse.ArgumentParser(description='Network and Port Scanner')
53    parser.add_argument('ip_range', help='IP range or subnet to scan for live hosts (e.g., 192.168.1.0/24)')
54    args = parser.parse_args()
55
56    # Step 1: Detect live hosts
57    live_hosts = detect_live_hosts(args.ip_range)
58    if not live_hosts:
59        print("No live hosts found.")
60        return
61
62    print(f"Live hosts detected: {live_hosts}")
63
64    # Step 2: Scan ports concurrently for live hosts
65    port_scan_results = scan_ports_concurrently(live_hosts)
66
67    # Step 3: Display the results
68    for host, open_ports in port_scan_results.items():
69        if open_ports:
70            print(f"{host} has open ports: {open_ports}")
71        else:
72            print(f"{host} has no open ports.")
73
74    # Optionally, check HTTP ports (80/443) using requests
75    for host in live_hosts:
76        http_ports = check_http_ports(host)
77        if http_ports:
78            print(f"{host} has open HTTP/HTTPS ports: {http_ports}")
79
80 if __name__ == "__main__":
81     main()
82

```



```
1
import arpaname

PS C:\Users\KRISHNA AHAR\Documents\1Stop Projects\1Stop Project2\Network & Port Scanner> .\venv\Scripts\activate
(venv) PS C:\Users\KRISHNA AHAR\Documents\1Stop Projects\1Stop Project2\Network & Port Scanner> python scanner.py 192.0.2.1
Scanning for live hosts in range: 192.0.2.1
No live hosts found.
(venv) PS C:\Users\KRISHNA AHAR\Documents\1Stop Projects\1Stop Project2\Network & Port Scanner> python scanner.py 176.32.103.205
Scanning for live hosts in range: 176.32.103.205
No live hosts found.
(venv) PS C:\Users\KRISHNA AHAR\Documents\1Stop Projects\1Stop Project2\Network & Port Scanner> python scanner.py 8.8.8.8
Scanning for live hosts in range: 8.8.8.8
Live hosts detected: ['8.8.8.8']
Scanning ports for 8.8.8.8
8.8.8.8 has open ports: [53, 443, 853]
(venv) PS C:\Users\KRISHNA AHAR\Documents\1Stop Projects\1Stop Project2\Network & Port Scanner> python scanner.py 192.168.1.0/24
Scanning for live hosts in range: 192.168.1.0/24
Live hosts detected: ['192.168.1.1', '192.168.1.2', '192.168.1.3', '192.168.1.5']
Scanning ports for 192.168.1.1
Scanning ports for 192.168.1.2
Scanning ports for 192.168.1.3
Scanning ports for 192.168.1.5
192.168.1.1 has open ports: [53, 80, 443]
192.168.1.2 has no open ports.
192.168.1.3 has no open ports.
192.168.1.5 has open ports: [135, 139, 445, 902, 912]
192.168.1.1 has open HTTP/HTTPS ports: [80]
(venv) PS C:\Users\KRISHNA AHAR\Documents\1Stop Projects\1Stop Project2\Network & Port Scanner> clear
```